

3 Many neurones in the human nervous system are myelinated.

(a) A myelinated axon conducts impulses faster than a non-myelinated axon of the same diameter.

Explain this difference.

(3)

(b) Polyneuropathy is a disorder that damages the myelin sheath of neurones throughout the body.

One symptom is muscle weakness. Muscle weakness is the reduced strength in one or more muscles.

Suggest how damage to the myelin sheaths of neurones can lead to muscle weakness.

(2)

- (c) Dementia is a condition associated with the ongoing decline of brain functioning. There are many types of dementia.

The relationship between myelin in brain tissue and types of dementia has been investigated.

The mean quantity of myelin in samples of brain tissue from groups of people with types of dementia and from a control group was measured.

The table shows the results of this investigation.

Group	Mean quantity of myelin in a brain tissue sample / a.u.	Standard deviation
Control (no dementia)	52	± 3.2
Patients with vascular dementia	25	± 5.9
Patients with Alzheimer's dementia	32	± 4.1
Patients with Lewy Body dementia	42	± 5.0

A student concluded that there was a relationship between the quantity of myelin in the brain of a person and whether or not they had dementia.

Comment on the validity of this conclusion.

(2)

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(Total for Question 3 = 7 marks)